

RF PCB Design Fundamentals

From microstrip to antenna — designing for the GHz world

Impedance • Microstrip • Antennas • Shielding

Audience: Intermediate Engineering Students

Z_0 • Substrate • Antenna keep-out • EMI/EMC overlap

Why RF is Different

Wires become antennas; layout becomes a circuit

Wavelength shrinks

2.4 GHz $\rightarrow \lambda \approx 12.5$ cm.
Trace lengths matter.

Impedance everywhere

Mismatched $Z_0 \rightarrow$ reflections,
lost power, antenna detunes.

Parasitics dominate

Pad capacitance, via inductance
change resonant frequency.

Substrate Dk matters

FR-4 $\neq$ Rogers $\neq$ Teflon.
Governs wavelength on PCB.

Shielding required

Cans + ground stitching
are not optional past 1 GHz.

Test equipment expensive

VNA, spectrum analyzer
required for proper validation.

Common RF Bands & Wavelengths

Know the frequencies you'll meet in practice



Band	Frequency	λ (free space)	Common use
LoRa / ISM	433 / 868 / 915 MHz	32-69 cm	Long-range IoT
GSM / 4G LTE	700-2700 MHz	11-43 cm	Cellular
Wi-Fi 2.4G	2.4-2.5 GHz	12 cm	Wi-Fi b/g/n
BLE / Wi-Fi 5G	5-5.8 GHz	5-6 cm	Wi-Fi 5/6
UWB	6.5-9 GHz	3-4 cm	UWB ranging
5G mmWave	24-40 GHz	7-12 mm	5G NR FR2
Bluetooth LE	2.4 GHz	12 cm	BLE 4/5
NFC	13.56 MHz	22 m	Tap-to-pay

Substrate Materials

Dielectric constant decides everything

Material	Dk (typ)	Loss (Df)	Use
FR-4 std	4.2-4.7	0.02	Up to ~3 GHz, cheap
FR-4 high-Tg	4.3-4.5	0.015	Better stability
Megtron 6	3.7	0.005	10G+ digital, mmWave
Rogers RO4350B	3.48	0.0037	Sub-6 GHz RF
Rogers RO3003	3.0	0.0013	24-77 GHz radar
PTFE / Teflon	2.1-2.3	0.0004	High-power RF, very pricey
Polyimide flex	3.2-3.5	0.008	Flex RF, antennas

FR-4 limit

Standard FR-4 is OK up to ~3 GHz. Anything past that, your losses dominate — switch to Rogers / Megtron.

Microstrip vs Stripline

Two common transmission line geometries

Microstrip vs Stripline

Microstrip (surface)

- ▶ Trace on outer layer above GND plane
- ▶ Easy to access for tuning
- ▶ Radiates if uncovered
- ▶ Higher loss than stripline
- ▶ Width set by stackup → 50 Ω target
- ▶ Tools: KiCad PCB calculator, Saturn PCB

Stripline (buried)

- ▶ Trace between two GND planes
- ▶ Self-shielded — low EMI
- ▶ Lower loss
- ▶ Cannot be probed easily
- ▶ Requires multilayer (4+ layers)
- ▶ Used in high-density boards

Antenna Design & Keep-Out

Where the RF actually leaves your board

- ▶ Most projects use a pre-certified module (ESP32, nRF52, RM-USB) — antenna included
- ▶ Discrete chip antenna (e.g. Johanson 2450AT18A100E) needs a keep-out 'island'
- ▶ No copper, no ground, no signals under or behind the antenna
- ▶ Minimum keep-out width: vendor-specific (often ~5 mm)
- ▶ Antenna feed line: 50 Ω microstrip with matching network (π or T)
- ▶ Always include matching network pads, even if you don't populate at first
- ▶ Final tuning happens with a VNA — adjust caps / inductors on the bench
- ▶ PCB antenna (loop, meander, IFA) requires a careful trace + ground island

Impedance Matching

From IC pin to antenna — match every transition

Why match?

Mismatch reflects power
back into the PA — wasted W.

L-match

$L + C = 2$ components.
Narrowband, simple.

π -match

Two C + one L = T.
Wider band, more tunable.

T-match

Two L + one C.
Good for high-Z to low-Z.

Balun

Differential $\rightarrow$ single-ended
IC output $\rightarrow 50\ \Omega$ antenna.

Tools

Smith chart, NanoVNA,
SimSmith software.

Module-Based Design — The Beginner Path

Pre-certified radios with all the hard work done

- ▶ ESP32-WROOM, nRF52840 modules, SX1276/8 modules, BG95 cellular modules
- ▶ FCC / CE / BIS / ARIB pre-certification included → much cheaper compliance
- ▶ Antenna integrated or attached via U.FL connector
- ▶ Module data sheet specifies copper keep-out, layout, decoupling
- ▶ Trade-off: higher unit cost, but no RF expertise required
- ▶ Most start-ups ship Rev 1 with a module, then re-design with chip-down for cost
- ▶ Common modules: ESP32-WROOM, RAK4631, SIMCom 7600, u-blox NINA

Shielding & Cans

Metal enclosure around the radio section

- ▶ Soldered metal can over the RF section
- ▶ Fence around the cavity (or cavity walls of the can)
- ▶ Ground stitching vias every $\lambda/20$ around the perimeter
- ▶ Reduces both emissions OUT and noise IN
- ▶ Required for many RF certifications above 1 GHz
- ▶ Removable cans for rework — but harder to seal

Stitching rule

Via spacing $< \lambda/20$ for the highest frequency you care about.
5 GHz Wi-Fi: $\lambda/20 \approx 3$ mm.

Compromise

Even without a metal can, a continuous ground pour around RF traces + stitching helps a lot.

RF Layout Do / Don't

Patterns that work vs that don't

DO: ground stitching

Vias every $\lambda/20$ around RF traces.

DO: 50 Ω microstrip

Width per stackup; matched.

DO: antenna keep-out

No copper near antenna island.

DON'T: split RF GND

Reference plane must be solid.

DON'T: long PA trace

Place PA right next to antenna.

DON'T: digital nearby

Clock harmonics into RX.

Common Beginner RF Mistakes

What the spectrum analyzer reveals

- ▶ Copper under the antenna — detunes it badly
- ▶ Wrong trace width for 50 Ω — vendor's recommended layout ignored
- ▶ Antenna feed line too long (more loss than expected)
- ▶ No ground stitching → resonances in the cavity
- ▶ Sharing AVDD with digital MCU — phase noise on the radio
- ▶ No matching network pads — can't tune in production
- ▶ Missing TVS on antenna line — ESD damages the PA
- ▶ Copper-to-edge clearance too small — antenna pattern distorts

RF Certification

Every product transmitting RF needs it

Region	Standard / Body	Typical cost (₹)	Pre-cert module saves you
India	WPC / BIS	1-3 L	Most of the cost
USA	FCC	3-8 L	Reduces to ID approval only
Europe	CE-RED	3-10 L	Reduces by ~70%
China	SRRC	2-5 L	Module pre-cert helps
Japan	MIC / Giteki	3-7 L	Module saves big
Korea	KC	2-5 L	Module saves big
Worldwide	Anatel, IC, etc.	varies	Cumulative if global product

RF Test Equipment

Bench vs lab

NanoVNA (₹3-5k)

Pocket VNA, 1-6 GHz.
Great for matching networks.

USB Spectrum (₹15-30k)

RTL-SDR, HackRF —
find out-of-band emissions.

Bench VNA (₹3-10L)

Keysight / Rohde — full S-params.
For production design.

Spectrum analyzer

Real-time RF emission view.
Pre-compliance.

Anechoic chamber

OTA (Over-the-Air) testing.
Final certification.

Field probe set

Near-field probes localise
EMI sources.

RF Best Practices

Habits of every senior RF engineer

- ▶ Start with a pre-certified module unless you have RF expertise
- ▶ Use vendor's exact reference layout — don't deviate
- ▶ 50 Ω microstrip from IC to antenna with matching network
- ▶ Continuous ground plane under all RF signals
- ▶ Antenna keep-out: no copper, no signals
- ▶ Ground stitching vias around RF section
- ▶ Separate LDO for RF supply
- ▶ TVS on antenna line
- ▶ Reserve matching network pads even if not populated
- ▶ Plan for certification cost early in the project

Key Takeaways

Key takeaways from this lecture

Modules first

Pre-cert radio = fastest path.

Z_0 matters

Microstrip = 50 Ω end-to-end.

Substrate Dk drives layout

FR-4 limit ~3 GHz.

Antenna island sacred

Keep-out + ground rules.

Shielding & stitching

Cavity-like RF section.

Tune with VNA

Final matching is iterative.

Recommended Reading

RF resources for self-study

- ▶ Joseph Carr — 'Practical Antenna Handbook' (intuitive intro)
- ▶ Pozar — 'Microwave Engineering' (graduate-level standard)
- ▶ RF Cafe online — calculators, articles, reference tables
- ▶ Texas Instruments — 'RF for Dummies' app note
- ▶ Nordic Semiconductor app notes — nRF52 antenna design
- ▶ Espressif — ESP32 RF layout guide (free download)
- ▶ Phil's Lab YouTube — practical RF PCB design walkthroughs
- ▶ Smith chart software: SimSmith, Smith v4.1 (free)

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ Joseph Carr — Practical Antenna Handbook
- ▶ Pozar — Microwave Engineering
- ▶ RF Cafe — online RF calculators & reference
- ▶ TI / Nordic / Silicon Labs RF app notes
- ▶ Espressif ESP32 / ESP-IDF RF tuning guide
- ▶ Phil's Lab YouTube — practical walkthroughs
- ▶ NanoVNA hardware + software (free)
- ▶ Smith chart tools: SimSmith, NanoVNA-Saver

Thank you • Build something this week.